Raian Latif Nabil

530-979-6571 mnabil@ucdavis.edu LinkedIn GitHub Portfolio

Education

University of California, Davis Sep 2023 – Present
Ph.D. in Computer Science — Advisor: Dr. Cindy Rubio-González *Davis, CA*

Bangladesh University of Engineering & Technology Feb 2016 – Feb 2021
Bachelor of Science in Computer Science & Engineering — Advisor: Dr. M. Saifur Rahman *Dhaka, Bangladesh*

Research Interest

Software Engineering, Programming Language, AI4SE

Research Experience

University of California, Davis Sep 2023 – Present
Graduate Student Researcher *Davis, CA*

- Developed **CI-Bench**, a framework for evaluating LLM-based automated program repair tools on real-world Continuous Integration (CI) failures.
  * Integrated three LLM-based repair agents (**SWE-Agent, Agentless, and AutoCodeRover**) to diagnose CI failures and generate candidate patches.
  * Adapted **BugSwarm**, a dataset of reproducible fail-pass CI job pairs, to support automated patch generation and validation.
  * Evaluated repair effectiveness on **100 real-world CI failures**, achieving up to **35% correct patches**.
- Developed **LLM4Build**, an LLM-driven framework for reproducing historical GitHub Actions CI builds across multiple programming languages.
- Developing **NumBugBench**, a benchmark and automated reproduction framework for real-world numerical bugs in scientific computing software.
- Contributed to the development and maintenance of **BugSwarm**, an infrastructure for mining and reproducing real-world CI failures.

Bangladesh University of Engineering & Technology May 2019 – Feb 2021
Undergraduate Researcher *Dhaka, Bangladesh*

- Developed **CD-MAWS**, an alignment-free phylogenetic estimation method using cosine distance on Minimal Absent Word (MAW) sets.
- Implemented and evaluated multiple alignment-based methods for phylogenetic analysis of DNA sequences.

Teaching Experience

University of California, Davis Jan 2026 – Mar 2026
Graduate Teaching Assistant *Davis, CA*

- **ECS 122B – Algorithm Design & Analysis:** Led discussion and lab sessions, conducted office hours, and graded assignments.

Professional Experience

OpenRefactory Inc. Jul 2021 – Jul 2023
Software Engineer *Santa Clara, CA*

- Contributed to **Intelligent Code Repair (iCR)**, a static analysis-based platform for automated bug detection and repair.
- Developed static analysis algorithms to detect security vulnerabilities in Java, Python and Go applications.
- Enhanced **CI/CD integrations** of iCR with GitHub, GitLab, and Bitbucket, supporting both cloud and enterprise environments.
- Contributed to cloud-native deployment infrastructure using **AWS, Azure, and Kubernetes** to support scalable operation.

Publications

NumBugBench: Mining and Reproducing Real-World Numerical Bugs
Raian Latif Nabil, Cindy Rubio-González
Under Review

LLM4Build: Leveraging LLMs to Reproduce GitHub Actions
Raian Latif Nabil, Hao-Nan Zhu, Cindy Rubio-González
Under Review

CI-Bench: Evaluating LLM-based Program Repair Tools on CI Failure Data
Raian Latif Nabil, Hao-Nan Zhu, Cindy Rubio-González
ICSE 2026, Demonstrations Track [link]

CD-MAWS: An Alignment-free Phylogeny Estimation Method Using Cosine Distance on Minimal Absent Word Sets
Naser Anjum, **Raian Latif Nabil**, Rakibul Islam Rafi, Shamsuzzaman Bayzid, M. Saifur Rahman
IEEE/ACM Transactions on Computational Biology and Bioinformatics, 2023 [link]

Honors & Awards

Graduate Fellowship, University of California, Davis *Fall 2023, Fall 2026*

Travel Grant, ICSE 2026, Rio de Janeiro, Brazil *2026*

Academic Service & Professional Activities

Student Volunteer, SPLASH/ISSTA 2026, Oakland, CA *2026*

Conference Co-Reviewer: ICSE 2024, PLDI 2024, ICSE 2025

Conference Attendance: FSE 2023, ASE 2024, ICSE 2026, SPLASH/ISSTA 2026